Raship Khadka
Civil Engineer
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Research Profile

Civil Engineer with interests in Water Resources Engineering, Water Supply Engineering, Hydrology, Irrigation Engineering, GIS, and climate-resilient infrastructure. Experienced in rural water supply design, watershed and springshed management, GIS-based analysis, and reservoir identification for water-induced disaster management.

Education

Bachelor of Civil Engineering
Institute of Engineering (IOE), Thapathali Campus, Tribhuvan University
2019–2024 Aggregate: 70.40/100
IELTS: 7.5
Relevant Coursework: Hydrology, Hydraulics, Water Supply Engineering, Irrigation Engineering, Hydropower Engineering, Fluid Mechanics, Computational Techniques, Numerical Methods.

Research Interests

Water Resources Engineering • Water Supply Engineering • Hydrology • Irrigation Engineering • Watershed & Springshed Management • GIS & Remote Sensing • Climate-Resilient Water Infrastructure

Professional Experience

Surveyor and Designer 2024–Present
Hints Consult Pvt. Ltd. | Kaustuv Engineering Consultancy Pvt. Ltd. | Development Activist Nepal Pvt. Ltd.

- Planned and designed rural and semi-urban water supply systems, including hydraulic analysis and distribution network design.
- Conducted watershed and springshed investigations for sustainable water resource management.
- Performed engineering surveys, hydraulic design, quantity estimation, and cost estimation for civil infrastructure projects.
- Contributed to bridge engineering projects through site surveys, technical documentation, and engineering design support.
- Conducted feasibility studies for micro-hydropower projects (up to 5 MW), including catchment assessment, hydrological analysis, discharge estimation, head calculation, preliminary layout planning, and technical feasibility evaluation.

Freelance Civil Design Engineer 2025–Present
Residential Building Design (Freelance)

- Conducting structural analysis and reinforced concrete design for a residential building, including the design of foundations, beams, columns, and slabs in compliance with relevant codes.
- Preparing structural drawings, reinforcement detailing, and construction documentation, while coordinating with the client to ensure functional and code-compliant design solutions.

Lecturer (Part-Time) May 2024–Sep 2025
School of Geomatics

Subjects Taught: Quantity Estimation • Land Development & CAD • Applied Mechanics • Engineering Drawing I & II • Computer Applications

Selected Research & Engineering Projects

Reservoir Identification and Water-Induced Disaster Management

- Conducted GIS-based reservoir site identification and suitability analysis for water-induced disaster management.

- Performed watershed delineation, hydrological analysis, reservoir storage estimation, and terrain assessment to support integrated water resources planning.

Rural Water Supply Projects

- Designed rural and semi-urban water supply systems, including population projection, demand forecasting, hydraulic analysis, and distribution network design.
- Prepared engineering drawings, quantity estimates, and technical reports for sustainable water supply infrastructure.

Residential Building Design (Freelance)

- Preparing architectural and structural drawings for a reinforced concrete residential building using AutoCAD.
- Developing structural layouts and construction drawings while coordinating with the client to meet design requirements.

Micro-Hydropower Feasibility Study

- Conducted site reconnaissance, hydrological assessment, discharge estimation, and preliminary layout planning for micro-hydropower projects.

Bridge Engineering Project

- Assisted in engineering surveys, technical documentation, quantity estimation, and design support for bridge infrastructure projects.

Publication

Hada, S., & Khadka, R. (2024). *Navigating Road Safety: Factors, Legislations, Safety Audits, Technological Advances and Challenges in Nepal.* KTFT Journal, 4(1), 78–92.
DOI: 10.3126/ktftj.v4i1.70364

Technical Competencies

Geospatial Analysis: ArcGIS Pro, QGIS
Hydraulic Engineering: HEC-RAS, HEC-HMS, EPANET, WaterGEMS
Structural Engineering: ETABS, SAFE
CAD/BIM: AutoCAD, Revit
Programming: Python, MATLAB
Technical Writing: Microsoft Office, LaTeX (Overleaf)

Leadership

Class Representative; Organizing Committee Member (National Tech Fest); Paper Bridge Competition Coordinator

Honors

Best Final Year Project Presentation • Mix Design Competition Winner • Nepal Scout Member

Languages

English • Nepali • Hindi

References

Er. Sukriti Suvedi
Assistant Professor, IOE
ssuvedi@tcioe.edu.np

Er. Rakesh Kumar Chaudhary
Ph.D. Student, Utah State University
logtokumarrakesh063@gmail.com